

CSE428 Section 01 & 02
Summer 2026 [MUNR]
Assignment 1

Instructions:

- The submission will be handwritten.
- Write your Name, Student ID and Section in the cover page.
- Naming convention: "StudentID_A1", e.g.:
13101001_A1
- Submit the PDF in this Google Form:
To be updated.
- Deadline: 8th August, 2026 (11.59PM)

Please make sure to submit your assignment only through the Google Form. Submissions elsewhere or after the deadline will not be accepted.

Question 1

Consider the following dataset:

Living Area (feet ²)	#Bedrooms	Price (X1000\$)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	516

- Using the dataset provided, estimate the parameters of a linear regression model to predict house prices based on the living area and the number of bedrooms. Starting from the given initial parameters, perform exactly **two iterations of gradient descent, showing the predicted values, cost, gradients, and updated parameters after each iteration**. Consider that, initial weights & biases are 0 and learning rate is 0.000001.
- Use the model from (a) to determine the price of a house that features 4 Bedrooms and 2200 square feet of Living Area.

Question 2

Consider the following dataset:

Living Area (feet ²)	#Bedrooms	Price Level
2104	3	Medium
1600	3	Low
2400	3	Medium
1416	2	Low
3000	4	High

- Using the dataset provided, train a logistic regression model to classify the house price level based on the living area and the number of bedrooms. Starting from the given initial parameters, perform exactly **two iterations of gradient descent**. **Show the predicted probabilities, binary cross-entropy loss, gradients, and updated parameter values after each iteration**. Consider that, initial weights & biases are 0 and learning rate is 0.000001.
- Use the model from (a) to determine the level of price of a house that features 4 Bedrooms and 2200 square feet of Living Area.

Question 3

- Why is cross-entropy loss used for classification tasks instead of mean-squared error? **Explain**.
- Differentiate** between overfitting and underfitting.
- What is regularization? **Differentiate** between L1 and L2 regularization.
- Briefly explain** various types of Gradient Descent.